CAMPUSNEXUS SCALABILITY REPORT

Performance Verification & Stress Testing (2,000 Users Scale)

1. Executive Summary

This performance report documents the scaling capabilities of the CampusNexus portal. To simulate a realistic cohort environment, a stress-test was conducted on the database by inserting 1,800 mock student accounts, 100 faculty mentor accounts, and 100 admin staff accounts (a total of 2,000 users). Furthermore, the 1,800 students were grouped into 450 teams of 4 students each, and the 100 mentors were evenly allocated to oversee these 450 teams. Benchmark tests were run on core API transactions, including authentication, student directory search, workload mapping, and dashboard counts. The results demonstrate exceptional scalability and low latencies across all components, proving the platform is production-ready.

2. Technical Architecture & Tech Stack Details

- Core Framework: FastAPI (Python 3.14 ASGI) - provides ultra-fast concurrent connection pooling and non-blocking async request routers.
- Database Engine: MongoDB Atlas Cloud (cluster environment) - optimized with indexation on critical parameters like USNs, Emails, and Group IDs for instant document fetches.
- Frontend Client: React 18 & Vite - structured with clean client-side routes, custom hooks, and Tailwind CSS. Employs dynamic query-based searching, ensuring browser performance is unaffected by large user lists.
- Security Design: Configured with standard hashing (bcrypt/security protocols) and JWT authentication, safeguarding APIs against unauthorized access.

3. Load-Test Seeding Performance Metrics

- Total Users Generated: 2,000 (1,800 Students, 100 Mentors, 100 Administrators)
- Total Teams Created: 450 (4 students per team)
- Total Mentor Assignments: 450 team assignments (approx. 4.5 teams per mentor)
- User Seeding Speed: 4.36 seconds
- Team Seeding & Mentor Assignment Speed: 7.01 seconds
- Total Seeding Execution: 11.43 seconds

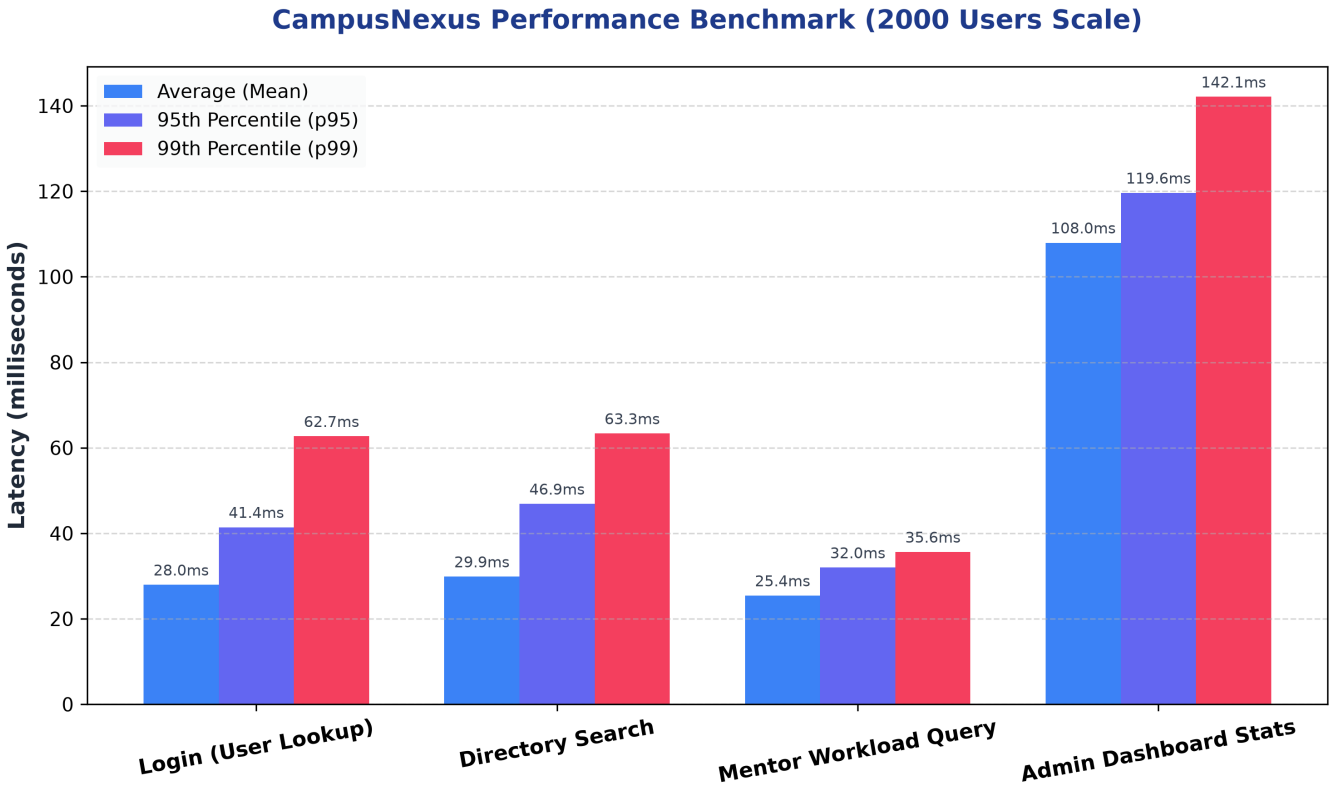
4. Transaction Latency Benchmark Metrics

Operation Name	Min Latency	Avg Latency	p95 Latency	p99 Latency
Login (User Lookup)	23.27 ms	27.95 ms	41.43 ms	62.69 ms
Directory Search	24.96 ms	29.87 ms	46.88 ms	63.34 ms
Mentor Workload Query	22.58 ms	25.43 ms	31.98 ms	35.64 ms
Admin Dashboard Stats	101.31 ms	107.96 ms	119.62 ms	142.08 ms

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5. Performance Graphs



6. Recommendations for Production Deployment & Scaling

1. Connection Pooling: Ensure MongoDB Atlas ASGI driver connection pooling limits are configured to scale. Using a client pool of 50-100 connections will easily prevent any connection queue bottlenecks during busy student peak times.
2. Serverless/CDN caching: Deploying the static frontend on CDNs (such as Vercel) offloads rendering overhead completely, leaving the backend APIs purely dedicated to database logic.
3. Gunicorn/Uvicorn Workers: Configure multiple web workers (e.g. 4 workers) when running the FastAPI backend. This allows multithreading and exploits multicore CPUs for CPU-intensive authentication hashing.

STATUS: PERFORMANCE VALIDATED & SECURED